

Øving 18

$$1) \quad x = [x], \quad F = [0], \quad G = [\sqrt{q}]$$

$$\underline{k=0:} \quad x_0 = [0] \quad P_0 = [1] \text{ m}^2$$

$$a) \quad \phi = I + F \Delta t = [1] + [0] \Delta t = \underline{\underline{[1]}}$$

$$b) \quad Q = \int_0^{\Delta t} \phi G G^T \phi^T d\Delta t = \int_0^{\Delta t} [1] [\sqrt{q}] [\sqrt{q}] [1] d\Delta t = \int_0^{\Delta t} q d\Delta t = \underline{\underline{[q \Delta t]}}$$

$$c) \quad q = 0.5 \frac{\text{m}^2}{\text{s}} \\ \Delta t = 10 \text{ s}$$

$$\begin{aligned} \tilde{x}_1 &= \phi_0 x_0 \\ &= [1][0] \\ &= \underline{\underline{[0] \text{ m}}} \end{aligned}$$

$$\begin{aligned} \tilde{P}_1 &= \phi_0 P_0 \phi_0^T + Q_0 \\ &= [1][1][1] + [5] \\ &= \underline{\underline{6 \text{ m}^2}} \end{aligned}$$

$$d) \quad \underline{k=1:}$$

$$\begin{aligned} z_1 &= 1 \text{ m} \\ \sigma_z &= 2 \text{ m} \end{aligned}$$

$$\begin{aligned} H_1 &= 1 \\ R_1 &= \sigma_z^2 = 2^2 = 4 \end{aligned}$$

Gain:

$$\begin{aligned} K &= \tilde{P}_1 H_1^T (H_1 \tilde{P}_1 H_1^T + R_1)^{-1} \\ &= 6 \cdot 1 (1 \cdot 6 \cdot 1 + 4)^{-1} \\ &= 6 \cdot \frac{1}{10} \\ &= \underline{\underline{0.6}} \end{aligned}$$

$$\begin{aligned} \hat{x}_1 &= \tilde{x}_1 + K(z_1 - H_1 \tilde{x}_1) \\ &= [0] + 0.6(1 - 0.6(0)) \\ &= 0 + 0.6 \\ &= \underline{\underline{0.6 \text{ m}}} \end{aligned}$$

$$\begin{aligned} \hat{P}_1 &= (I - KH) \tilde{P}_1 \\ &= (1 - 0.6(1)) 6 \\ &= (0.4) 6 \\ &= \underline{\underline{2.4 \text{ m}^2}} \end{aligned}$$

$$2) \quad x = \begin{bmatrix} x \\ \dot{x} \end{bmatrix} \quad F = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad G = \begin{bmatrix} 0 \\ \sqrt{q} \end{bmatrix}$$

$k=0$:

$$x_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ m/s} \quad P_0 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \text{ m}^2/\text{s}$$

$$a) \quad \phi = I + F \Delta t$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \Delta t$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & \Delta t \\ 0 & 0 \end{bmatrix}$$

$$= \underline{\underline{\begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}}}$$

$$b) \quad Q = \int_0^{\Delta t} \phi G G^T \phi^T d\Delta t = \int_0^{\Delta t} \left(\begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ \sqrt{q} \end{bmatrix} \begin{bmatrix} 0 & \sqrt{q} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \Delta t & 1 \end{bmatrix} \right) d\Delta t = \int_0^{\Delta t} \begin{bmatrix} \sqrt{q} \Delta t \\ \sqrt{q} \end{bmatrix} \begin{bmatrix} 0 & \sqrt{q} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \Delta t & 1 \end{bmatrix} d\Delta t$$

$$= \int_0^{\Delta t} \begin{bmatrix} 0 & q \Delta t \\ 0 & q \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \Delta t & 1 \end{bmatrix} d\Delta t = \int_0^{\Delta t} \begin{bmatrix} q \Delta t^2 & q \Delta t \\ q \Delta t & q \end{bmatrix} d\Delta t = \underline{\underline{q \begin{bmatrix} \frac{1}{3} \Delta t^3 & \frac{1}{2} \Delta t^2 \\ \frac{1}{2} \Delta t^2 & \Delta t \end{bmatrix} \text{ m}^2/\text{s}^2}}$$

$$c) \quad q = 0.01 \frac{\text{m}^2}{\text{s}^3}$$

$$\Delta t = 10 \text{ s}$$

$$\hat{x}_1 = \phi_0 \hat{x}_0 = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0+10 \\ 0+1 \end{bmatrix}$$

$$\underline{\underline{\hat{x}_1 = \begin{bmatrix} 10 \\ 1 \end{bmatrix} \text{ m/s}}}$$

$$\tilde{P}_1 = \phi_0 \tilde{P}_0 \phi_0^T + Q_0$$

$$= \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 10 & 1 \end{bmatrix} + Q_0$$

$$= \begin{bmatrix} 0 & 10 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 10 & 1 \end{bmatrix} + 0.01 \begin{bmatrix} \frac{(10)^3}{3} & \frac{(10)^2}{2} \\ \frac{(10)^2}{2} & 10 \end{bmatrix}$$

$$= \begin{bmatrix} 100 & 10 \\ 10 & 1 \end{bmatrix} + \begin{bmatrix} 3.333 & 0.5 \\ 0.5 & 0.1 \end{bmatrix}$$

$$\underline{\underline{\tilde{P}_1 = \begin{bmatrix} 103.333 & 10.5 \\ 10.5 & 1.1 \end{bmatrix} \text{ m}^2/\text{s}^2}}}$$

d) $k=1$:

$$z_1 = 12 \text{ m}$$

$$H_1 = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$\sigma_z = 2 \text{ m}$$

$$R_1 = \sigma_z^2 = (2 \text{ m})^2 = [4] \text{ m}^2$$

Gain:

$$K = \tilde{P}_k H_k^T (H_k \tilde{P}_k H_k^T + R_k)^{-1}$$

$$K = \begin{bmatrix} 103.333 \\ 10.5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \left(\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 103.333 & 10.5 \\ 10.5 & 1.1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + [4] \right)^{-1}$$

$$= \begin{bmatrix} 103.333 \\ 10.5 \end{bmatrix} (103.333 + 4)^{-1}$$

$$K = \begin{bmatrix} 103.333 \\ 10.5 \end{bmatrix} \cdot (107.333)^{-1} = \underline{\underline{\begin{bmatrix} 0.9627 \\ 0.0978 \end{bmatrix}}}$$

$$\hat{x}_1 = \tilde{x}_1 + K(z_1 - H_1 \tilde{x}_1)$$

$$= \begin{bmatrix} 10 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.9627 \\ 0.0978 \end{bmatrix} (12 - \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 10 \\ 1 \end{bmatrix})$$

$$= \begin{bmatrix} 10 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.9627 \\ 0.0978 \end{bmatrix} (2)$$

$$= \begin{bmatrix} 10 \\ 1 \end{bmatrix} + \begin{bmatrix} 1.9254 \\ 0.1956 \end{bmatrix} = \underline{\underline{\begin{bmatrix} 11.9254 \\ 1.1956 \end{bmatrix} \text{ m/s}}}$$

$$\hat{P}_1 = (I - K_1 H_1) \tilde{P}_1$$

$$= \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.9627 \\ 0.0978 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \right) \begin{bmatrix} 103.333 & 10.5 \\ 10.5 & 1.1 \end{bmatrix}$$

$$= \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.9627 & 0 \\ 0.0978 & 0 \end{bmatrix} \right) \begin{bmatrix} 103.333 & 10.5 \\ 10.5 & 1.1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.0373 & 0 \\ -0.0978 & 1 \end{bmatrix} \begin{bmatrix} 103.333 & 10.5 \\ 10.5 & 1.1 \end{bmatrix}$$

$$= \begin{bmatrix} 3.85 & 0.39 \\ 0.39 & 0.07 \end{bmatrix} \text{ m}^2/\text{s}^2$$